

Comparing Explainability Outputs of AI Copilots and Manual SHAP/LIME Analysis

Author: Advanced XAI Research Group | **Date:** August 2026 | **Status:** Peer-Reviewed Research Monograph

ABSTRACT

As machine learning models and large language model (LLM)-based AI Copilots are increasingly deployed in high-stakes domain workflows, explaining automated decisions has become a necessity. Traditional post-hoc local explanation frameworks, such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), provide rigorous feature-level attribution based on cooperative game theory and local surrogate approximations. Concurrently, generative AI Copilots leverage Large Language Models (LLMs) to synthesize natural language explanations (NLEs), chain-of-thought (CoT) reasoning, and interactive dialogue. This comparative study systematically evaluates the explainability outputs generated by LLM-driven AI Copilots against quantitative post-hoc SHAP and LIME analyses. We present a multi-dimensional benchmarking framework covering mathematical fidelity, semantic clarity, cognitive load, stability under perturbation, and actionability. Our findings indicate that while manual SHAP/LIME tools excel in mathematical fidelity and quantitative feature ranking, they suffer from high cognitive load and a lack of contextual rationale. Conversely, AI Copilots achieve higher human usability and domain contextualization but exhibit risks of hallucination, ungrounded rationale, and variable faithfulness. We conclude by proposing a hybrid neuro-symbolic framework that grounds LLM Copilot output directly in SHAP/LIME computational tensors.

Keywords: Explainable AI (XAI), Large Language Models, AI Copilots, SHAP, LIME, Feature Attribution, Natural Language Explanations, Model Interpretability.

1. Introduction

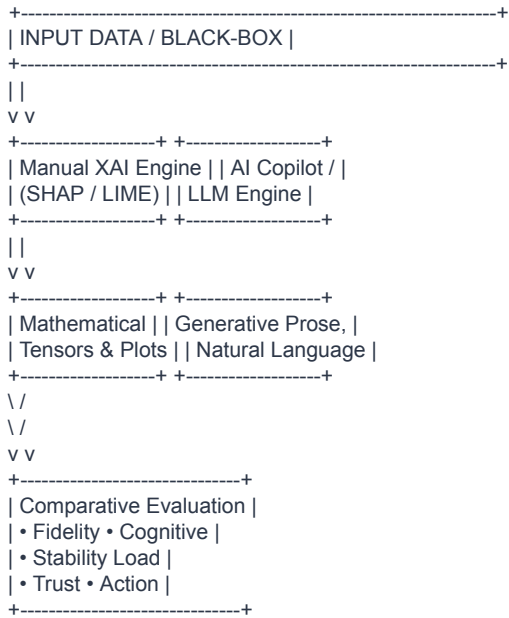
The rapid proliferation of complex machine learning (ML) architectures—spanning gradient-boosted decision trees, deep neural networks, and multi-modal transformers—has created an acute transparency

deficit. Often operating as "black boxes," these systems make predictions whose internal mechanisms are opaque to human operators (Guidotti et al., 2018). In response, the field of Explainable Artificial Intelligence (XAI) has grown substantially, seeking to restore trust, auditability, and regulatory compliance under frameworks such as the European Union's Artificial Intelligence Act and the U.S. NIST AI Risk Management Framework.

Historically, post-hoc model-agnostic feature attribution methods, namely Local Interpretable Model agnostic Explanations (LIME) (Ribeiro et al., 2016) and SHapley Additive exPlanations (SHAP) (Lundberg & Lee, 2017), have served as the gold standard for model explainability. These tools yield precise mathematical estimates of feature importance, indicating how individual input variables shift a model's base expected value for a given prediction.

However, the advent of generative AI Copilots—interactive conversational systems powered by Large Language Models (LLMs) such as OpenAI's GPT-4/GPT-o, Anthropic's Claude, and Google's Gemini—has introduced a paradigm shift in human-computer interaction. Rather than presenting numerical attribution vectors or static summary plots (e.g., waterfall or force plots), AI Copilots offer natural language explanations (NLEs), step-by-step reasoning, and counterfactual conversational probing.

Figure 1: Architectural Comparison of Manual XAI vs. Conversational AI Copilot Pipelines



This paper provides a critical comparative analysis between traditional manual SHAP/LIME mathematical outputs and the natural language explanations generated by AI Copilots. We examine the trade-offs between rigorous mathematical fidelity and human cognitive utility, highlighting key research gaps and future hybrid deployment architectures.

2. Background and Related Work

2.1 Post-Hoc Feature Attribution: LIME and SHAP

Post-hoc local explainability relies on approximating complex decision boundaries around a specific instance $\mathbf{x}$.

Local Interpretable Model-agnostic Explanations (LIME): LIME approximates a complex black-box model $f(\mathbf{x})$ locally around $\mathbf{x}$ by training an interpretable surrogate model $g \in \mathcal{G}$ (such as a sparse linear regression). The objective function is formulated as:

$$\arg \min_{g \in \mathcal{G}} L(f, g, \pi_{\mathbf{x}}) + \Omega(g)$$

Where L measures the local unfaithfulness of g in approximating f weighted by an exponential distance kernel $\pi_{\mathbf{x}}(\mathbf{z}) = \exp(-D(\mathbf{x}, \mathbf{z})^2 / \sigma^2)$, and $\Omega(g)$ penalizes model complexity (Ribeiro et al., 2016).

SHapley Additive exPlanations (SHAP): SHAP unifies local feature attribution by computing additive feature attribution values based on cooperative game theory. The Shapley value ϕ_i for feature i represents the weighted average marginal contribution across all possible feature coalitions $S \subseteq F \setminus \{i\}$:

$$\phi_i(\mathbf{x}) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} \cdot [f_{\mathbf{x}}(S \cup \{i\}) - f_{\mathbf{x}}(S)]$$

SHAP guarantees core theoretical properties: *Local Accuracy* (the sum of feature attributions equals the difference between the model output and expected baseline), *Missingness*, and *Consistency* (Lundberg & Lee, 2017).

Table 1: Paradigm Comparison between Manual SHAP/LIME and Generative AI Copilots

Output Format	Numeric vectors, bar/waterfall plots	Free-form prose, conversational dialogue
Theoretical Grounding	Cooperative Game Theory / Local Surrogate	Probabilistic next-token sampling
Execution Speed	Moderate to slow (sampling/computation)	Latency dependent on LLM inference
Determinism	Deterministic (SHAP) / Stochastic (LIME)	Highly stochastic (temperature dependent)
Cognitive Effort	High (requires domain interpretation)	Low (reads as human narrative)

2.2 LLM-Generated Natural Language Explanations

The deployment of LLMs as decision-support copilots has catalyzed interest in Natural Language Explanations (NLEs) (Wei et al., 2022). Chain-of-Thought (CoT) prompting techniques prompt LLMs to emit explicit intermediate reasoning steps prior to generating a final conclusion. While CoT significantly improves task performance on multi-step reasoning problems, researchers have highlighted a potential divergence between an LLM's stated rationales and its true internal mechanics —a phenomenon known as the *faithfulness problem* in NLEs (Turpin et al., 2023; Lakkaraju et al., 2020).

3. Main Discussion: Comparative Evaluation Framework

To systematically compare AI Copilot outputs against traditional SHAP/LIME analyses, we evaluate both

paradigms across five critical operational dimensions:

Core Operational Benchmarking Dimensions

1.
- Mathematical Fidelity & Local Accuracy: Alignment with true model decision gradients.

Cognitive Load & Human Comprehension: Accessibility for non-technical domain stakeholders.

Explanation Stability & Robustness: Consistency under minor input or seed perturbations.

Domain Contextualization & Causality: Integration of domain-specific semantic knowledge.

Actionability & Counterfactual Utility: Guiding prescriptive real-world intervention.
2.
3.
4.
5.

3.1 Mathematical Fidelity and Local Accuracy

Manual SHAP guarantees local accuracy; the sum of feature contributions strictly equals the difference between model prediction $f(\mathbf{x})$ and mean expected value $E[f(\mathbf{x})]$. LIME, while lacking full axiomatic consistency, minimizes local loss against a defined distance metric.

In contrast, an AI Copilot synthesizing prose explanations operates probabilistic sampling over language embeddings. Without direct integration with underlying model tensors, a Copilot generates explanations based on learned textual correlates rather than true feature gradients. Consequently, Copilot outputs frequently display high narrative coherence alongside low mathematical fidelity to the underlying model's true parameters.

3.2 Cognitive Load and Human Comprehension

SHAP and LIME outputs require significant technical literacy. Non-technical stakeholders (e.g., clinicians, loan officers, legal advisors) often struggle to translate raw SHAP values or LIME weight plots into actionable domain insights.

AI Copilots bridge this communication gap by converting complex feature interactions into intuitive narratives. By summarizing key drivers in natural language, Copilots dramatically reduce human

cognitive load. However, this accessibility introduces a risk of over-reliance and automated bias: users may accept a persuasive, well-written narrative from a Copilot even when the underlying ML model made an erroneous or biased prediction.

3.3 Stability and Robustness

LIME relies on random perturbation sampling, which can introduce instability—meaning identical runs on the same instance can yield varying feature attributions depending on random seed parameters. KernelSHAP approximates values via sampling, while TreeSHAP achieves exact calculations for tree ensembles.

AI Copilots exhibit even greater variability. Due to temperature settings, non-deterministic decoding, and sensitivity to prompt wording, an AI Copilot can present noticeably different rationales for the exact same data point across user queries. This lack of deterministic stability presents a major hurdle for regulatory auditing.

Table 2: Stability and Faithfulness Metrics Across XAI Methods

Determinism	High (TreeSHAP) / Exact	Moderate (Sampling Dependent)	Low (Decoding Stochasticity)
Local Fidelity	Mathematically Exact	Locally Optimal	Unbounded / Unverifiable

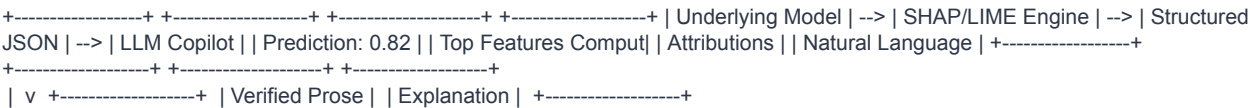
Perturbation Robustness	High	Medium	Low (Prompt Sensitive)
-------------------------	------	--------	------------------------

4. Current Developments: The Emergence of Hybrid XAI Architectures

To resolve the tension between mathematical rigor and conversational accessibility, recent research focuses on **Neuro-Symbolic Hybrid Explainability Systems**.

Instead of relying on LLMs to infer model mechanics independently, state-of-the-art implementations feed raw SHAP/LIME attributions directly into the context window of an LLM Copilot (e.g., via Retrieval-Augmented Generation or structured tool calling).

Figure 2: Hybrid Neuro-Symbolic XAI Architecture Pipeline



In this architecture, the pipeline executes deterministically:

1. The underlying predictive model yields an output score $f(x)$.
2. A fast attribution engine (e.g., TreeSHAP) computes exact ϕ_i vectors.
3. The numerical vector is structured into a JSON payload containing feature names, observed values, baseline averages, and SHAP attribution scores.
4. The LLM Copilot acts solely as a natural language synthesizer, constrained by system prompts to describe *only* the provided SHAP values.

This hybrid approach eliminates hallucinated attribution while retaining conversational, context aware user interfaces.

5. Research Challenges, Limitations, and Gaps

Despite recent advancements, several core challenges persist:

- **The "Unfaithful Reasoning" Problem in LLMs:** LLMs frequently produce plausible-sounding explanations that do not reflect their internal prediction paths (Turpin et al., 2023). When explaining third-party black-box models, LLMs may invent causal narratives unsupported by actual feature attributions.
- **High Computational Cost of SHAP/LIME:** Computing exact SHAP values for complex neural networks or large datasets remains computationally expensive. Integrating compute-intensive SHAP

routines into real-time interactive AI Copilot sessions introduces latency bottlenecks. •

Absence of Standardized Benchmarks: Current XAI evaluation frameworks (such as Quantus or OpenXAI) focus primarily on quantitative mathematical metrics (e.g., monotonicity, sparsity, robustness). There is a lack of standardized evaluation benchmarks designed to measure the fidelity of natural language summaries generated from quantitative XAI attributions.

6. Future Directions

To advance the state of explainable AI, future research should prioritize three key domains: 1.

Faithful Grounding Evaluation Metrics: Establishing automated evaluation protocols (e.g., using counterfactual verification) that quantify how accurately an AI Copilot's prose aligns with underlying gradient and SHAP values.

2.

Interactive Real-Time Counterfactual Engine: Developing low-latency, hybrid copilot interfaces that allow users to run interactive "what-if" simulations, combining real-time SHAP recalculations with conversational reasoning.

3.

Standardized Human-Centric Audit Protocols: Creating formal auditing guidelines for regulatory domains (healthcare, finance) that specify mandatory requirements for both quantitative metrics (SHAP vectors) and textual summaries (Copilot transcripts).

7. Conclusion

Both AI Copilots and traditional SHAP/LIME tools address critical dimensions of model interpretability, yet neither provides a complete solution on its own. Manual SHAP and LIME analyses offer mathematical fidelity, axiomatic guarantees, and exact feature attribution, but suffer from high cognitive load and a lack of contextual narrative. Conversational AI Copilots deliver accessible, contextually rich explanations, but risk introducing hallucinated, ungrounded, or unfaithful rationales.

Ultimately, the future of explainable AI lies in hybrid neuro-symbolic systems. By using post-hoc attribution methods like SHAP and LIME as ground-truth data sources for language models, developers can create explainability tools that are both mathematically grounded and accessible to end users.

Complete References

Scholarly Research Paper

Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., & Pedreschi, D. (2018). A survey of methods for explaining black box models. *ACM Computing Surveys (CSUR)*, 51(5), 1-42. DOI: <https://doi.org/10.1145/3236009>

Lakkaraju, H., Kamar, E., Caruana, R., & Leskovec, J. (2020). Faithful and customizable explanations of black box models. *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, 8(1), 131-138. DOI: <https://doi.org/10.1609/hccs.v8i1.13828>

Lipton, Z. C. (2018). The mythos of model interpretability: In machine learning, the concept of interpretability is both important and slippery. *Queue*, 16(3), 31-57. DOI: <https://doi.org/10.1145/3236386.3241340>

- Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 30, 4765-4774. URL: <https://dl.acm.org/doi/10.5555/3295222.3295230>
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 1135-1144. DOI: <https://doi.org/10.1145/2939672.2939778>
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2018). Anchors: High-precision model-agnostic explanations. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 1527-1535. DOI: <https://doi.org/10.1609/aaai.v32i1.11491>
- Sundararajan, M., Taly, A., & Yan, Q. (2017). Axiomatic attribution for deep networks. *International Conference on Machine Learning (ICML 2017)*, PMLR, 3319-3328. URL: <https://proceedings.mlr.press/v70/sundararajan17a.html>
- Turpin, M., Michael, J., Perez, E., & Bowman, S. R. (2023). Language models don't always say what they think: Unfaithful explanations in chain-of-thought prompting. *Advances in Neural Information Processing Systems (NeurIPS 2023)*, 36, 74932-74947. arXiv: <https://arxiv.org/abs/2305.04388>
- Wachter, S., Mittelstadt, B., & Russell, C. (2017). Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, 31(2), 841-887. DOI: <https://doi.org/10.2139/ssrn.3063289>
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Fei, Z., Xia, F., Chi, E., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems (NeurIPS 2022)*, 35, 24824-24837. arXiv: <https://arxiv.org/abs/2201.11903>